LIAM GERON

Senior Data Science Specialist
Machine Learning + NLP+ LLM Systems

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Summary

Machine learning specialist with 6+ years of experience building production NLP, predictive modeling, and LLM systems across insurance, SaaS, and healthcare. Experienced designing end-to-end ML platforms—from large-scale data pipelines and feature engineering to retrieval-augmented LLM applications and enterprise MLOps infrastructure. Proven track recorddelivering measurable business impact including major improvements in sales conversion, fraud detection, and operational automation.

Core Expertise

Machine Learning: predictive modeling, churn analysis, customer segmentation, recommender systems

LLM & NLP: OpenAI / HuggingFace LLMs, embeddings, semantic search, text classification

ML Infrastructure: Databricks, PySpark, Spark SQL, Airflow, ETL pipelines

Deployment: Flask/FastAPI, Docker, AWS, GCP, Azure MLOps

Languages: Python, SQL, PySpark, R

Experience

Quility Insurance
Senior Data Science Specialist
August 2023 – Present

Design and deploy machine learning and NLP systems supporting sales optimization, fraud detection, and enterprise AI infrastructure.

• Built large-scale **sales conversation classification system** using OpenAI embeddings andlogistic regression to detect resistant vs. non-resistant leads across hundreds of thousands ofagent–lead messages.
• Applied **CorEx topic modeling** to categorize resistance types, enabling targeted outreach that increased conversion rates from **4% to 27%.**
• Automated processing of **hundreds of thousands of insurance application PDFs**, extracting structured PII for fraud detection models and analytics workflows.
• Designed **Azure-based MLOps pipeline** to package and deploy machine learning models asenterprise APIs, improving reliability and accessibility of predictive services.
• Architected **retrieval-augmented generation (RAG) LLM system** ingesting internal webpages into a vector database for semantic search and enterprise knowledge retrieval.

Cognizant
Senior Associate, Data Science
March 2022 – June 2023

Delivered machine learning and NLP systems for enterprise clients in automotive and customer analytics.

• Improved sentiment analysis model accuracy by **16%** using LLM embeddings and Flair models deployed via Airflow pipelines.
• Built **Random Forest lead conversion model** achieving **94% precision,** enabling salesteams to prioritize high-probability leads.
• Processed **100+ GB of raw enterprise data** using PySpark and Spark SQL to automatedashboards and ML pipelines used by business leadership.

Chatdesk
Data Scientist
March 2018 – April 2020

Developed machine learning systems for automated tagging and analysis of customer feedbackacross messaging platforms.

• Built backend ML pipeline processing **20M+ customer support messages** across **80+ companies** using Python, Flask, GCP, and AWS.
• Developed NLP classification models including logistic regression, neural networks, and GPT-based approaches.
• Generated analytics insights that improved **CSAT scores** and reduced ticket handling time for clients.

Education

CUNY Graduate Center
MA Computational Linguistics — GPA 4.0

New York University (NYU)
BA English